

Flight Test Plan and Feasibility Study Report

Autonomous Control Reusable Sounding System

Project Leader: Kong Yichen

Test Flight Time: 2026 Year July Early

2026 Year May

Abstract

This report proposes and demonstrates an intelligent autonomous control reusable sounding system based on high-altitude balloons. The system ascends vertically using a standard 750 g balloon, separates from the balloon after reaching the stratosphere, and achieves long-range precise recovery through powered gliding with variable-sweep wings. The report covers feasibility analysis and field experiment processes in areas such as system design, aerodynamic and structural simulation, avionics and control laws, flight test plans, and safety assurance. The conclusion

indicates that the scheme is technically feasible and has a high level of safety redundancy, making it suitable for conducting field test verification within the proposed time window.

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1 Project Background and Value

The current high-altitude meteorological radio sounding operations generally adopt a disposable non-recoverable model, facing long-term hardware financial consumption, ecological environmental pollution caused by debris falling, and transnational data security and legal risks brought about by the drift of high-altitude wind direction in border areas. In addition, traditional sounding instruments are suspended below large balloons, and during the ascent phase, they are often subjected to thermal disturbances and mechanical turbulence generated by the balloon's top, which frequently leads to systematic observational biases in temperature and humidity data.

This study proposes and implements a green, intelligent, and recyclable "autonomous control and recoverable exploration system based on high-altitude balloons." The system uses a single standard 750 g high-altitude balloon as the sole power source for the vertical ascent phase, towing a variant gliding probe with a highly innovative aerodynamic shape to ascend vertically. Upon reaching the preset stratospheric altitude (approximately 26,000 m ~

After completing the multi-factor profile detection at 30,000 m, the system achieves non-destructive cutting of the sphere through an electrically controlled release mechanism. The aircraft autonomously transitions into a powered gliding state with variable wing angles in the thin stratosphere, relying on self-tuning control laws to drive the fully movable tail control surfaces for high-precision three-axis decoupled navigation, ensuring a precise and safe return across borders within

a maximum radius of 300km. Finally, it utilizes a surface sled for a non-destructive landing, achieving dozens of cycles of reuse for expensive precision detection instruments.

1.1 Introduction to the Research Balloon

Nowadays, there are many methods for high-altitude meteorological observation, typical ones include sounding balloons, meteorological satellites, satellite occultation, meteorological radar, aircraft-mounted sensors, ground-based remote sensing, and rocket sounding. Meteorological satellites and satellite occultation technology can obtain observation profiles from top to bottom (in contrast to meteorological radar), and through methods such as weighting, the signal results from a certain layer often mix with signals from other layers, which cannot accurately reflect the true situation of a specific layer. A simple example is that it can severely underestimate the warming of the troposphere and cooling of the stratosphere caused by the greenhouse effect. Aircraft-mounted sensors and rocket sounding cannot stably obtain dense data points of the atmospheric vertical profile, especially for wind field detection, and are relatively expensive. Sounding balloons use sensors for direct contact observation, achieving a transition from two-dimensional to three-dimensional observation, which can represent the most authentic atmospheric conditions. As a direct detection method for high-altitude meteorology, sounding balloons also have advantages such as slow ascent, detailed data, and accurate measurement results, and are often used for calibrating other detection instruments.

The sounding instrument can measure atmospheric physical parameters at different heights in the sky to determine the vertical distribution of meteorological elements. The sounding instrument can be classified into general and specialized types based on its use. The sensors for meteorological elements in sounding instruments have undergone several transformations, including: temperature wires, temperature rods, thermistors, bimetallic gauges, humidity-sensitive resistors, humidity-sensitive capacitors, and silicon pressure gauges. Since traditional sounding instruments are single-use, the sensors are relatively simple. The measurements are all relative quantities, and before formal release, a "baseline determination" must be conducted to relate the sensor's relative values to the absolute values of the environment in which it is located, making it easier to process and convert the detection records later.

Currently, weather balloons play a crucial role in studying various physical processes in the atmosphere, weather analysis and forecasting, and various scientific meteorological services.



Figure 1: The 750g high-altitude balloon used in this test flight (not connected to the sounding instrument), taken by me at the high-altitude meteorological observation station.

1.2 Existing Issues with the Research Balloon

1.2.1 Research Balloon for Environmental Issues

1. The scale of use is continuously expanding: every year, hundreds of thousands of meteorological balloons are launched into the sky worldwide. These balloons are essential for collecting data on high-altitude temperature, pressure, and humidity, forming the foundation of modern weather forecasting. However, meteorological balloons also leave a harmful legacy in the world's oceans. While they are crucial for collecting data used in weather reports and storm warnings, they are virtually unrecoverable. Once they reach an altitude of about 30 kilometers, latex balloons expand and burst, scattering debris everywhere. These balloons are equipped with small radio sounding instruments. Due to high-altitude winds, most of this equipment falls into the ocean, and their debris poses a serious threat to marine life and the natural environment. Several countries have reported that debris from weather balloons has led to the deaths of albatrosses, sea turtles, and other animals, and this issue is becoming increasingly severe as the number of weather balloons used rises.

2. Environmental pollution is becoming more serious: weather balloons are equipped with various observation instruments such as parachutes, thermometers, hygrometers, and barometers made of metal materials, as well as batteries and circuit boards. They are packed into a foam polystyrene box measuring 10×10×15 centimeters; most materials degrade slowly in nature and pose significant hazards. For example:

- (1) Research conducted in Australia estimates that up to 70% of Australia's weather balloons eventually end up in the ocean. Over a period of 21 months, they collected more than 2,400 pieces of balloon debris around the Great Barrier Reef. This is just one example of a widespread problem in the marine environment.

- (2) The United States launches approximately 76,600 balloons each year. If the strings of these balloons were connected end to end, they could stretch to the summit of Mount Everest 185 times. About 25,000 of the high-altitude meteorological sounding balloons released by the Japan Meteorological Agency fall into the sea each year. Our country currently has about 123 high-altitude meteorological observation stations, with each sounding station launching 2 to 4 balloons daily.

Continuously throughout the year, it is conservatively estimated that each weather station releases only 1 sounding balloons per day, resulting in a total number of balloons launched i

a year of approximately 121-141-2-8979 balloons, and the actual situation is likely to be even more.

3. Diplomatic issues: Non-recoverable reconnaissance balloons can easily lead to diplomatic problems. Europe has coordinated multiple times on this issue, and the phenomenon of reconnaissance balloons crossing borders occurs frequently around the world.



Figure 2: The situation of the GTS1 digital sounder falling in the field

1.2.2 Issues with the sounding balloon itself

1. Cost issues are prominent: According to Zhou Heng, an inspector from the International Department of the China Meteorological Administration, the average price of sounding instruments used in daily meteorological operations internationally is around 1000 yuan. In contrast, the sounding instruments used for meteorological observations domestically are usually cheaper, costing over 500 yuan (the price of the Beidou sounding instrument, which is currently under verification, is around 1000 yuan). Additionally, the unit price of hydrogen balloons typically ranges from dozens to hundreds of yuan, and launching a sounding balloon costs about 600 yuan. With China having 120 sounding stations that launch balloons twice a day, 365 days a year, the total cost for launching sounding balloons alone amounts to 52.56 million yuan annually (based on traditional sounding devices), which is quite expensive for repetitive meteorological measurements.
2. Data transmission capability is limited: As the probe balloon is currently non-recoverable, its data relies on ground L-band radar for real-time reception. The new generation depends on the BeiDou short message communication function, and both have a small amount of information transmitted; there are significant restrictions on the launch location, and detecting in oceans and wilderness is challenging, often requiring open environments.
3. Poor stability:
 - (1) Currently, the inflation of high-altitude balloons is done manually, and field investigations have found that the amount of inflation by meteorological station personnel has a high degree of uncertainty. This leads to instability in the ascent rate and height for each measurement, and sometimes the required altitude of 26000m may not be reached, or the balloon may drift too far at excessive heights. A small number of areas have adopted automatic inflation and release systems, but they also face issues such as high costs and lack of portability.

(2) Traditional L-band radar executes sounding with the radar as the coordinate center, using radio ranging and tracking the sounding instrument to obtain

The continuously changing spatial coordinates are then used to infer wind direction and speed information. Due to the influence of radar performance, Earth's curvature, atmospheric refraction, and other factors, the accuracy of the spherical coordinates calculated by the radar is unstable.

(3) Traditional sounding devices are prone to losing balls, especially during the moment of release when the phenomena of "overhead, sidelobe, and ball loss" are severe. The new Beidou sounding

instrument also easily loses tracking when there are significant changes in altitude, often requiring the ball to be replayed.

(4) At this stage, some sensors of the sounding device are exposed, and during thunderstorms, lightning strikes are severe, which can easily cause balloon releases.

4. Complex operation: The traditional sounding device is relatively complex to operate, heavily reliant on manual labor, and often requires multiple people to collaborate. During field investigations, sounders often need to run back and forth between the inflation chamber, gas storage chamber, operation room, and balloon release area, which is time-consuming and labor-intensive.
5. Restrictions on outdoor scientific research: Currently, the use of sounding balloons in outdoor scientific research is increasing. When conducting outdoor research, additional sensors often need to be installed. In the past, scientific research often required custom-made instruments, which had long lead times and high costs.
6. Signal interference: Nowadays, especially in overseas regions such as Europe and the United States, sounding instruments face serious mixed threats and cyber threats, such as GNSS interference and radio attacks. For sounding instruments that primarily use radio signals for data transmission, the data is easily subject to intentional or unintentional interference, which may lead to distorted meteorological data and inaccurate weather forecasts. Typically, the data transmission of sounding instruments is not encrypted, making them vulnerable to mixed threats. Given the sensitivity of meteorological data, the security and reliability of data transmission are crucial, and addressing these threats is essential for maintaining the integrity and reliability of meteorological data.

1.3 Precedents of test flights for similar projects

There have been test flight precedents both domestically and internationally where aerial balloons lead to high-altitude recovery of autonomous flying devices, with a typical example being:

- DLR HABLEG project: In 2015, the German Aerospace Center (DLR) collaborated with several aerospace companies to conduct research on a compact unmanned flying platform named "High-Altitude Balloon Launch Experiment Glider" (HABLEG).

Structurally, the wingspan of the glider is 3 meters, with a weight of 7.4 kilograms. It is transported to an altitude of 19550m using a high-altitude balloon, released at an altitude of 266km above the ground, and autonomously returns to the base. The mission duration is 145 minutes, and the aircraft ultimately lands 80 meters from the launch point, testing the hardware, flight software, and scientific payload of future large high-altitude pseudo-satellite (HAPS) platforms at moderate cost, while accumulating practical operational experience for high-altitude and beyond-visual-range missions.

https://elib.dlr.de/100773/1/HABLEG_ESA_PAC_2015_paper.pdf



Figure 3: DLR HABLEG Project Flight Route Map and Test Area

- HORUS Project: On May 17, 2023, scientists from the National Oceanic and Atmospheric Administration (NOAA), CIRES, the University of Colorado, Arizona State University, Delta Zee Solutions, and Spektreworks jointly developed a system named HORUS (High-altitude Operational Return Unmanned System), which was carried by a high-altitude balloon to an altitude of 90,000 feet (approximately 27.4 kilometers) and returned to the launch site in Colorado with scientific instruments.

[https://research.noaa.](https://research.noaa.gov/revolutionary-noaa-high-altitude-research-tool-passes-key-milestone/)

[gov/revolutionary-noaa-high-altitude-research-tool-passes-key-milestone/](https://research.noaa.gov/revolutionary-noaa-high-altitude-research-tool-passes-key-milestone/)



Figure 4: HORUS Project Test Flight Preparation in Pawnee National Grassland, Colorado

- Reusable Multi-Purpose Sounding System: In 2025, the unmanned intelligent detection technology R&D team at the Institute of Atmospheric Physics, Chinese Academy of Sciences successfully tested a reusable sounding system. This system includes airborne and ground subsystems: the airborne subsystem mainly consists of a glider, sounding instruments, and balloons; the mobile ground subsystem's main functions are glider path planning, balloon deployment, and sounding data reception and flight trajectory monitoring.

Launch, sounding data reception, and flight trajectory monitoring. In this trial, a lightweight glider integrated with a sounding device was carried aloft by a balloon, released at altitudes of 10,000 meters and 17,000 meters, and flew against the wind at multiple levels, with the final landing point being less than 200 meters from the preset point, achieving rapid and damage-free recovery.

https://iap.cas.cn/gb/xwdt/zhxw/202507/t20250709_7883217.html



Figure 5: Successful test flight news from the Institute of Atmospheric Physics, Chinese Academy of Sciences

2 Brief description of the autonomous recoverable sounding system

This project has won first prize twice in the "Qinghai Province Youth Science and Technology Innovation Competition," and has been shortlisted for the "National Youth Science and Technology Innovation Competition" for two consecutive years, with a national invention patent application submitted.

The sounding system aims to achieve autonomous recovery of the sounding device after separation from the high-altitude balloon using a novel variant layout of the aircraft, employing variable-sweep wings to adapt to strong winds at high altitudes and maximize flight efficiency, which is innovative on a global scale. This system helps improve environmental pollution and biological threats, enhances detection safety, and addresses issues such as uncontrollable landing points, high funding consumption, payload capacity limitations, and data transmission capability constraints. It mainly includes a sounding reception station and a sounding device.



National Intellectual Property Administration

102299

4th Floor, Room B380, Building 14, Cuiwei Middle Lane,
Haidian District, Beijing Huajuan Yingsheng Patent Agency (General Partnership)
Shi Baozhong (17601029869)

Date of issuance:

May 8, 2026



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Patent application acceptance notice

According to Article 28 of the Patent Law and Articles 43 and 44 of its implementation rules, the patent application submitted by the applicant has been accepted by the National Intellectual Property Administration. The determined application number, application date, and

Application Number: 2026106259652

Application Date: May 8, 2026

Applicant: Xining Second Middle School

Inventor: Kong Yichen

Invention Name: Self-recyclable Sounding Device

Upon verification, the National Intellectual Property Administration confirms receipt of the following documents:

Claims Document: 1 copy, 1 page; Number of Claims: 10

Description: 1 copy, 4 pages

Drawings Attached to the Description: 1 copy, 10 pages

Abstract of the Description: 1 copy, 1 page

Power of Attorney for Patent Agency: 1 copy, 2 pages

Request for Invention Patent: 1 copy, 4 pages

Substantive Examination Request Document Count: 1 copy

Application Scheme Volume Number: 2605478

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Examiner: Automatic Acceptance
Contact Number: 010-62356655

Review Department: Preliminary Review and Process Management Department

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Figure(s): Patent Application Acceptance Notice

2.1

Innovation Points

This project involves aerospace technology and meteorological monitoring, with the following innovations:

- (1) It achieves the autonomous recovery of sounding balloons, including the recovery of a large amount of meteorological records, with the potential to improve environmental pollution, diplomatic conflicts, high funding consumption, and limited data transmission capabilities.
- (2)
- (3) During the descent phase of non-recoverable missions, wind measurement or vertical meteorological profile detection can be conducted, which can replace downward-sounding instruments for more accurate vertical meteorological profile detection.

It provides solutions for the handling of non-recoverable missions, as well as for structural components that have reached their service life or are damaged. All selectable structural components (such as main materials, sealants, adhesives, gas transmission devices, etc.) aim to make more parts of the device biodegradable and low-pollution, ensuring that its environmental friendliness is no worse than, or even better than, traditional sounding instruments. Expired or damaged devices can be recycled or degraded.

- (4) Many improvements have been made to the appearance of the sounding device, and multiple fluid analyses have been conducted to ensure that the turbulence generated by high-altitude balloons does not distort the sensors due to the removal of the connecting rope.
- (5) The radio frequency system operates in the ultra-high frequency band, with direct connections without intermediate relays, using the LoRa CSS modulation method instead of traditional GFSK and other modulation methods. The ground station employs a 10-unit high-gain Yagi antenna, while the sounding device uses a dipole antenna, both utilizing linear polarization to achieve low power, long-distance, wide bandwidth, and high robustness data transmission.
- (6) In extreme weather conditions or special situations (such as meteorological disasters), the sounding device is not recovered. After completing the ascent and hold tasks, the ball pressure is reduced again through an electromagnetic valve to implement the meteorological detection task during the descent phase (compared to recovery situations, additional data on wind speed and direction is collected), maximizing detection efficiency and fully utilizing individual detection capabilities.
- (7) A brand-new aerodynamic layout concept for aircraft: a novel variant aircraft with a blended body and variable-sweep single-wing layout, innovatively applying the swept wing to slow, small aircraft, significantly enhancing wind resistance and flight efficiency, while effectively utilizing its lateral and side coupling to reduce the turning radius.
- (8) The ends of the swept wing are equipped with blended wingtip sails (Horner type + swept type + wingtip sails), which provide excellent drag reduction effects, reducing induced drag at various angles and providing a certain amount of forward thrust, with minimal variation in drag across different angles.
- (9) A segmented internal and external dual-structure design, with high structural strength, making it easy to add and remove sensors.
- (10) The solenoid valve controls the ball pressure, with the high-altitude balloon directly connected to the sounding device. The built-in solenoid valve adjusts the ball pressure based on the set ascent speed and realizes retention detection, reducing operational difficulty.
- (11) Highly scalable, it can be paired with different sensors according to task requirements, and is easy to operate. The circuit board, payload compartment, and aerodynamic design have all been optimized, achieving integration of general and specialized sounding instruments under a ball weight of 750g.
- (12) The flight control algorithm adopts the MBD development method, reducing parameter tuning, and has general applicability for this type of sounding device.

2.2 Research Summary

This section provides a brief overview of the research content; for detailed research, please refer to the "Research Report".

2.2.1 Aerospace Design

A new type of aircraft with a variable-sweep wing layout and integrated fuselage has been designed. The wing shape has been improved using Xflow and Xflr5, and the entire aircraft has undergone CFD simulation using Fluent. The aircraft adjusts the sweep angle to maximize flight efficiency and wind resistance, with no control surfaces on the swept wing, relying on a fully movable tail with edge strips for control. The cross-coupled flight control algorithm is developed using the MBD method. The sounding device operates in an environment with significant variations in humidity and air pressure, and the entire structure adopts a segmented internal and external dual-structure design, following GB/T 3452 to design an IP66 waterproof structure. The connecting rope has been eliminated, with the high-altitude balloon directly connected to the sounding device, a one-way valve used for inflation, and an electromagnetic valve for exhaust and high-altitude retention. A fairing is designed to prevent turbulence from the high-altitude balloon from distorting the sensors. The prototype model weighs 560g, the minimum MCU system.

The final product is expected to weigh around 70g, meeting the load requirement of 70g for the balloon.

2.2.2 Circuit Design

Designed a four-layer main control PCB, using STM32H743VIT6 as the main control chip, Beitian BH-280 (to prevent magnetic sensor errors, a low-power model without a compass was selected), a four-mode GNSS module, and Weite Intelligent JY61P six-axis IMU as the attitude sensing unit. For the extremely cold environment during flight, a custom low-temperature lithium polymer battery pack was used, along with a purely hardware-based automatic power switching circuit and a high-current DC-DC voltage regulator, ensuring redundancy and stable power supply during flight.

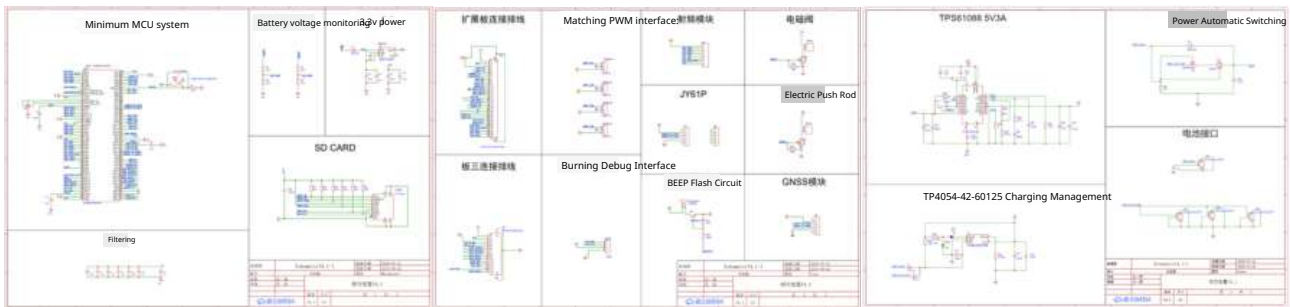


Figure7: Schematic

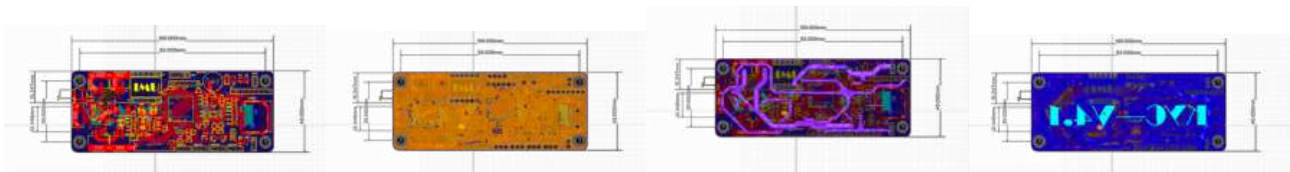


Figure8: Wiring Diagram

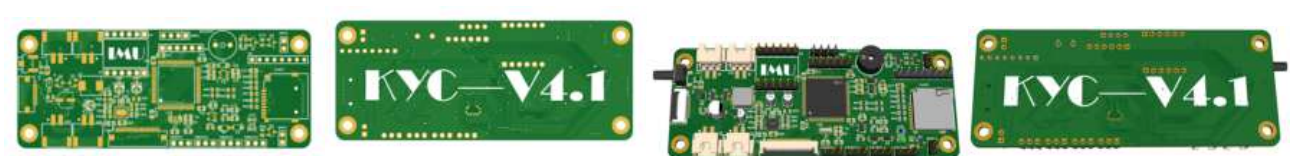


Figure9: Physical Image

2.2.3 Communication System Design

I have obtained an amateur radio operator certificate and applied for a license, qualifying me to legally use amateur radio equipment in the ultra-high frequency band. The communication system design employs LoRa CSS modulation method, with the ground station using 10 dBS of high-gain Yagi antennas+ a self-tracking gimbal, and the sounding device using a dipole antenna, all utilizing linear polarization to achieve low power, long distance, wide bandwidth, and high robustness data transmission. The equipment has completed CMA verification and certification at the Qinghai Province Radio Monitoring Station, with all parameters meeting the requirements.



Figure10: My Radio Station Operating Certificate and License



Sample Name: ds射模块

Commissioning Inspection

Client: Kong Yixie

Report Date: February 9, 2006



1 Notes

Without approval, no part or portion of this report may be reproduced.

If the commissioning unit has any comments on certain numbers and records in the testing services, please submit a written opinion to our site within 15 yuan. The validity period of the service is one year.

II. Testing Unit Information

Name: Qinghai Province Radio Monitoring Site
Address: 24 Wuyi Road, Chengzhong District, Xining City, Qinghai Province
邮编: 810000
Phone: 0971800609718467106
传真: 09718466037

III. Client Information

Client: Kong Yijia (Individual)
Mailing Address: Room 302, No. 1, West Road, Chengxi District, Xining City, Qinghai Province
Postal Code: 10000
Phone: 12997320997
Contact Person: Xuan Yikuan
Email: /

Radio Transceiver Equipment Testing Report

Radio Transceiver equipment testing kit			
产品名称	无线收发测试仪	型号规格	RF-2019-03200-04
制造商名称	无一人	联系人姓名	王正星
设备形式	便携式	制造商	010712300001
生产厂家	312031300000041	原产地	1221-4
使用地点	北京顺义区空港	品牌名称	RF11-10417
主要配置	20-2000MHz	重量	25.70kg
质保期	1年	交货日期	2020年2月8日

quality period of the service is one year.

the commissioning unit.

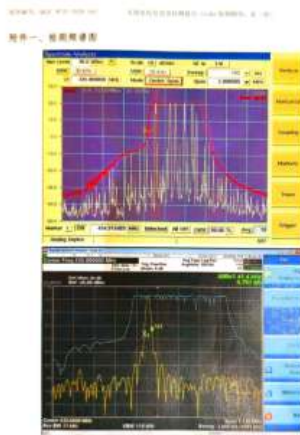
编写: 张介霖 审核: 司泽明 批准: 卢金良

職上 関・熱・水・煙

Testing Data Sheet

序号	检测项目	检测结果	标准值	判定结果
1	无丝织品	0	4000h	合格
2	最大允许裂缝宽度	0.18	1.75	合格
3	保护层厚度	0	41.88h	合格
4				

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0000-0000-0000-0000



(Must-buy power is very strong)

第五卷 第三十卷

Figure11: Radio transceiver equipment testing report

The communication system conducted ground tests in the Garzashika - Mensuan Basin, with the receiving end at an altitude of 3800m and the transmitting end at an altitude of 2700m. The tes

ion distance was approximately 775 nm, and the results for exceeded those of ordinary omnidirectional monopole antennas and disc antennas

2.3 Manufacturing

The sounding system has currently produced a prototype, primarily using PMI (Polymethyl methacrylate) rigid foam m + HT-PLA-GF (glass fiber reinforced high-temperature polylactic acid composite material), and has conducted feasibility tests for balloon inflation and long-distance communication. The final processing method and manufacturer have been selected. If on-site launch is feasible, production, manufacturing, and assembly debugging will take place in early June. This section provides a brief description of the selected manufacturing materials, processes, and manufacturers.



Figure12: Prototype model

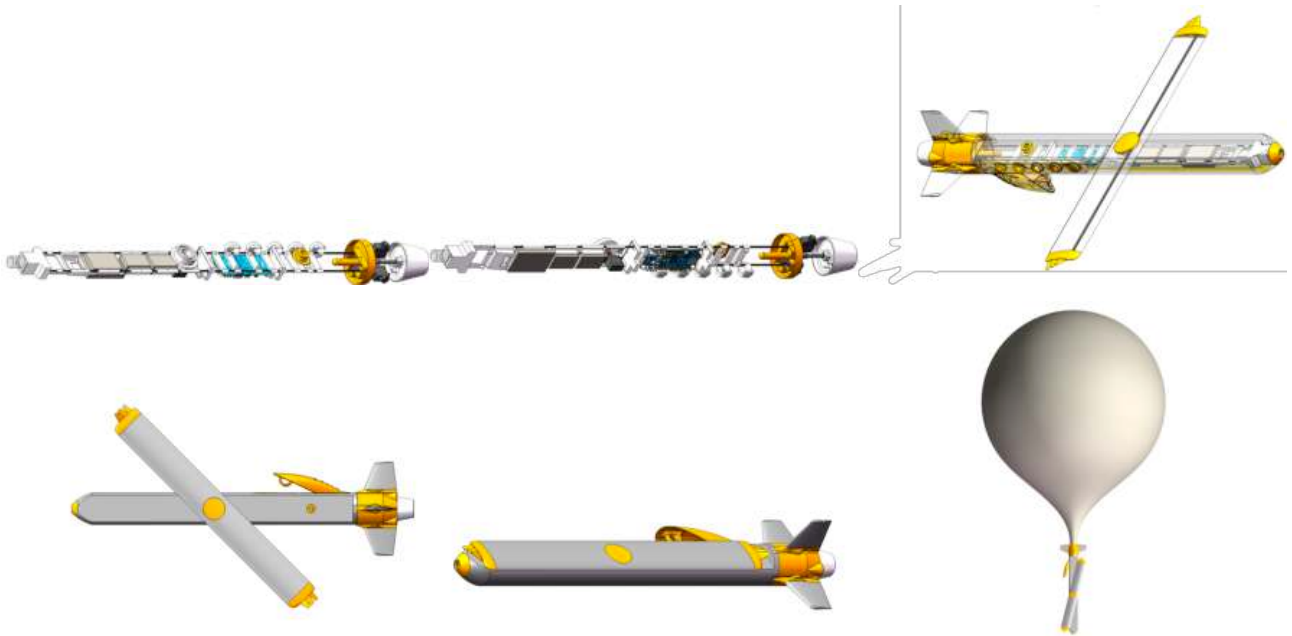


Figure13: 3D modeling

1. Wing main body: PMI foam core material and carbon fiber composite sandwich, using $75 \pm 7 \text{ kg/m}^3$ of XK75 PMI, CNC processing, with the composite process bei

g vacuum bag curing. The main characteristics are summarized as follows:

- (1) Mechanics and dimensions: Meet the required compressive strength and modulus for design, with a compressive strength of approximately 1.1 kN and a tensile modulus of approxim

ately 80 MPa.

- (2) Temperature and medium resistance: high heat distortion temperature, resistant to various lubricating oils, fuels and solvent effects.

- (3) Processing and bonding: Easy to CNC process, can reliably bond with epoxy resin, low moisture absorption, and low dielectric loss.

2. The body shell: The hollow shell of polyamide 11 (PA11) based on SAF technology mainly undertakes

pneumatic resistance barrier, component accommodation, and extreme cold thermal protection functions. Given the extreme low temperature below -50°C at an altitude of 30,000 meters.

Table1: PM1 structural core material properties

Item	CK40	CK52	CK72	CK110	XK32	XK52	XK75	XK110	XK200	WK32	WK52	WK75	WK110
Density(kg/m ³)	40±5	52±5	72±7	110±11	32±5	52±5	75±7	110±11	200±20	32±5	52±5	75±7	110±11
Compressive Strength(MPa)	≥0.55	≥0.65	≥1.20	≥2.00	≥0.29	≥0.80	≥1.50	≥2.30	≥6.40	≥0.29	≥0.80	≥1.50	≥2.30
Compression modulus(MPa)	≥28	≥32	≥40	≥60	≥12	≥30	≥40	≥80	≥160	≥12	≥30	≥40	≥80
130°C Compressive Strength(MPa)	≥0.35	≥0.40	≥0.80	≥1.15	≥0.20	≥0.40	≥0.80	≥1.20	≥2.20	≥0.20	≥0.40	≥0.80	≥1.20
Tensile Strength(MPa)	≥0.40	≥0.60	≥1.60	≥2.40	≥0.30	≥1.10	≥1.70	≥2.70	≥5.80	≥0.60	≥1.10	≥1.70	≥2.70
Tensile Modulus(MPa)	≥30	≥30	≥78	≥100	≥25	≥50	≥80	≥140	≥290	≥25	≥50	≥80	≥140
Shear Strength(MPa)	≥0.40	≥0.50	≥1.30	≥1.40	≥0.25	≥0.50	≥1.30	≥1.80	≥3.00	≥0.25	≥0.50	≥1.30	≥1.80
Shear Modulus(MPa)	≥12	≥14	≥24	≥35	≥8	≥14	≥24	≥40	≥100	≥8	≥14	≥24	≥40
Porosity(%)	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00	≤5.00
Dimensional Change Rate(%)	≤1.00	≤1.00	≤1.00	≤1.00	≤1.00	≤1.00	≤0.90	≤0.85	≤0.75	≤1.00	≤1.00	≤1.00	≤1.00
Heat Distortion Temperature(°C)	≥180	≥180	≥180	≥180	≥180	≥180	≥180	≥180	≥180	≥180	≥180	≥180	≥180
Compression Strength After Medium Resistance(MPa) — 3# Aviation Kerosene/ 15# Hydraulic Oil/ 4109 Lubricating Oil/ Acetone/ 70°C 4109 Lubricating Oil													
3# Aviation Kerosene	≥0.55	≥0.65	≥1.20	≥1.80	≥0.29	≥0.50	≥1.20	≥2.30	≥6.40	≥0.29	≥0.80	≥1.50	≥2.30
15# Hydraulic Oil	≥0.55	≥0.65	≥1.20	≥1.80	≥0.29	≥0.50	≥1.20	≥2.30	≥6.40	≥0.29	≥0.80	≥1.50	≥2.30
4109 Lubricating Oil	≥0.55	≥0.65	≥1.20	≥1.80	≥0.29	≥0.50	≥1.20	≥2.30	≥6.40	≥0.29	≥0.80	≥1.50	≥2.30
Acetone	≥0.55	≥0.65	≥1.20	≥1.80	≥0.29	≥0.50	≥1.20	≥2.30	≥6.40	≥0.29	≥0.80	≥1.50	≥2.30
70°C 4109 Lubricating Oil	≥0.55	≥0.65	≥1.20	≥1.80	≥0.29	≥0.50	≥1.20	≥2.30	≥6.40	≥0.29	≥0.80	≥1.50	≥2.30

At low temperatures, conventional engineering plastics (such as PLA and ABS) can suffer from severe low-temperature brittleness, leading to structural failure. Therefore, the shell is made from polyamide 11 (PA11) engineering plastic, which has top-notch low-temperature impact resistance. Parameters indicate that this material has a tensile strength of 51 MPa and a fracture elongation rate of 30%, ensuring that the structure maintains high flexibility and elastic deformation capability even at the extremely cold top of the troposphere. The manufacturing process employs Selective Absorption Fusion (SAF) technology, which utilizes precise spraying of high-absorption fluids combined with infrared heat sources for unidirectional uniform heating and melting. Its components exhibit extremely high and uniform isotropy in three-dimensional space, with high interlayer strength in the Z-axis, effectively preventing the thin-walled hollow shell from delaminating under external torsional forces. The shell wall thickness is set at 1.0 mm, with an integrated inner wall constructed with evenly spaced annular reinforcement frames, effectively resisting the dynamic pressure of high-speed airflow under lightweight constraints, and providing non-destructive energy absorption protection for the internal core avionics during landing recovery.



Introduction

PA11Pro, due to its higher ductility, stronger impact resistance, and broader adaptability for industrial applications, is suitable for mass production of robust and precise components. Additionally, PA11Pro is made from castor oil, featuring 100% bio-based characteristics, making it an environmental

Material Advantages

- High ductility
- Stronger impact resistance
- Environmentally friendly material

Precision

$\pm 300 \mu\text{m}$ or $\pm 0.3\%$

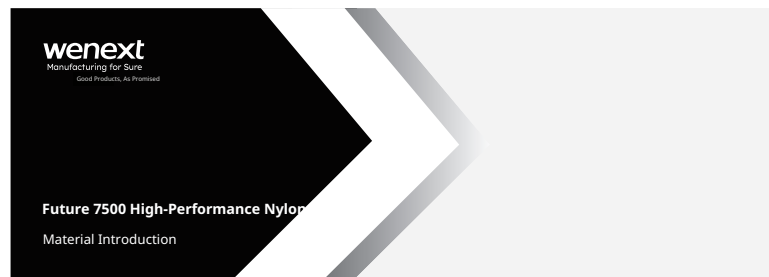
Performance Parameters

Heat deformation temperature (0.45MPa, 1.82MPa): 185°C, 47°C
Tensile modulus (ASTM D 638): 1.53GPa
Tensile Strength (ASTM D 638): 51MPa
Elongation at Break (ASTM D 638): 30%
Flexural Modulus (ASTM D 790): 1.34GPa
Flexural Strength (ASTM D 790): 51MPa
Notched Impact Strength(XZ,YX)(ASTM D256): 7.4KJ/m ²
Notched Impact Strength(ZX)(ASTM D256): 4.5KJ/m ²

Material testing shows no cytotoxic effects.
Skin irritation tests showed no irritation.
The above are the test results of the raw materials. If the product's post-processing involves painting or dyeing, it m

Figure 14: PA11 Material Properties

3. Structural Stress Components: The precision components made of polyamide 12 (PA12) based on the MJF process are core structural stress parts such as the installation seat for the variable wing servo inside the fuselage, mechanical linkage adapters, and main control fixed frames. High rigidity, low creep, and rigid limits are the primary design indicators. These components use high-performance polyamide 12 (PA12) material and are manufactured using the Multi Jet Fusion (MJF) process. The MJF process, with its micron-level droplet control of the fusion agent and refining agent, achieves extremely high geometric boundary sharpness and feature size molding accuracy. The PA12 components produced by this process have a bending modulus of 1800 MPa and a bending strength of 70 MPa. Their high rigidity characteristics ensure that the variable wing drive mechanism does not experience elastic play under multi-axis composite loads, eliminating the risk of mechanical jamming caused by strong winds. At the same time, the extremely high microscopic density of MJF components provides good mechanical sealing and thermal insulation, effectively slowing down the heat absorption of internal batteries and sensors from strong convection at high altitudes when combined with localized thermal insulation design.



Introduction

The Future 7500 High-Performance Nylon (HP3DHR-PA12) is the first material launched in the market with the introduction of HP industrial-grade 3D printing equipment. The HP-3D printer uses a jet fusion 3D printing solution, achieving extremely high dimensional accuracy and exquisite detail expression through HP's unique multi-material printing process, producing parts with the ideal mechanical properties and

Material Advantages

With exquisite detail expression and ultra-high dimensional accuracy;

Can produce highly precise small holes, thin walls, rod handles, sturdy structural components, complex parts, and lattice structures.

Precision

Below 100mm, $\pm 300\mu\text{m}$

Above 100mm, $\pm 0.3\%$

Performance Parameters

Heat Deflection Temperature (ASTM D 648 @66PSI): 155°C
Melting Point: 187°C
Tensile Modulus (ASTM D 638): 1800 MPa
Tensile Strength (ASTM D 638): 48 MPa
Elongation at Break (ASTM D 638): 15-20%
Flexural Strength (ASTM D 790): 70 MPa
Flexural Modulus (ASTM D 790): 1800 MPa
Notched Impact Strength (ASTM D 256): 3.5-3.6
Coefficient of Thermal Expansion, °C (TMA(T<Tg)): --
Poisson's Ratio (ASTM D 638): --
Dielectric constant 60 Hz (ASTM D 150-98):--

Future 7500 High-Performance Nylon

Figure1.5: PA12 Material Properties

- Non-structural load-bearing components: Conventional fused deposition modeling (FDM) lightweight auxiliary components for non-structural load-bearing components that do not participate in the main load path of the entire machine, but only serve as spatial separators, wiring guides, physical limits for batteries, and temporary tooling functions, are manufactured using conventional fused deposition modeling (FDM) processes. Lightweight modified polymers (such as foamed LW-PLA or PETG) are selected as materials. Since these components bear minimal load, the design focus is entirely on reducing volume and enabling rapid iteration verification. In modeling, a single-layer outer wall shell is used in conjunction with low-density honeycomb (such as Gyroid structure) internal filling (with a filling rate strictly controlled at 3% to 5%), aiming to maximize the reduction of the auxiliary component's mass quota, completely allocating the excess weight budget to the wings and main variable wing mechanisms.
- Core load-bearing skeleton: Pre-impregnated carbon fiber plate CNC machined main longitudinal beam serves as the core area where all loads are concentrated. The central rotating axis of the variable wing, high-torque drive servos, and the bending moments and shear forces transmitted from the wings on both sides exhibit highly localized stress concentration characteristics, which cannot be maintained solely by polymer components. Therefore, a "main keel" is designed inside the fuselage, which is made of 2.0mm thick high-performance industrial-grade pre-impregnated carbon fiber plate. This plate is

prefabricated using standardized hot-pressing techniques, achieving a density close to 100% and extremely high isotropic mechanical properties. Two-dimensional longitudinal main load-bearing plates are processed through CNC wire cutting, and the mechanical rotating shaft and actuating servos are rigidly locked onto the carbon fiber plate using metal fasteners. This "bone and flesh separation" design mechanism completely confines peak stress and torsional bending moments within the carbon fiber longitudinal beam and achieves self-dissipation, thoroughly avoiding the risk of cracking in the outer plastic shell due to overload.

2.4 Overview of Usage and Flight Procedures

Since this sounding device is developed based on existing sounding platforms, and the 750g high-altitude balloon, inflation equipment, and launch site used are consistent with traditional sounding instruments or Beidou sounding instruments, it can directly perform gas inflation and launch operations at existing high-altitude meteorological observation stations.

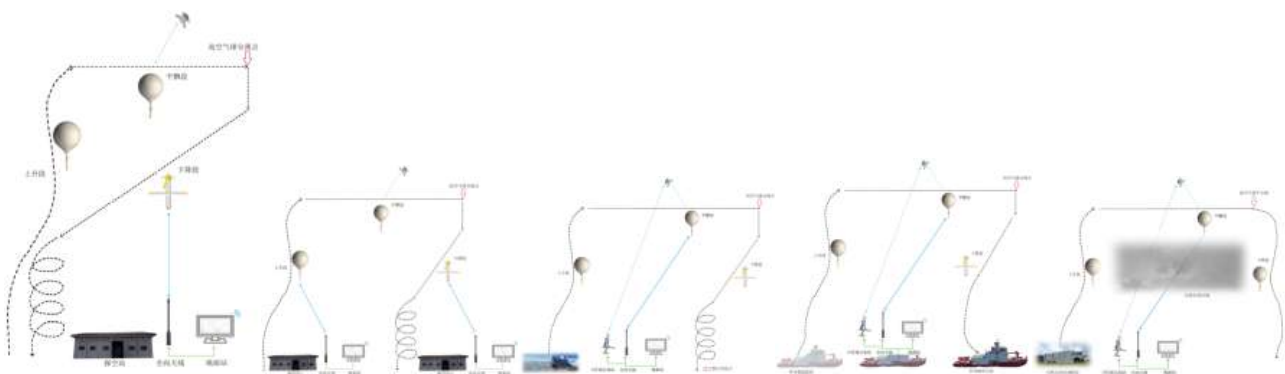


Figure 16: 5 Task Modes, this test flight verifies the basic mode on the left

The implementation process and specific steps of this device are as follows:

Step1 : Replace or confirm the payload.

Step2 : Performance self-check and charging.

Step3 : Sensor baseline measurement.

Step4 : Connection of the high-altitude balloon and intelligent inflation control; at this point, bidirectional radio communication is established, and data transmission begins.

Step5 : Execution of ascent phase detection tasks and ascent speed control.

Step6 : Optional horizontal hovering detection at specific altitude layers.

Step7 : Continue detection or autonomous recovery; based on preset or real-time instructions, the recovery sounding device can be released from the high-altitude balloon, with the option to continue detection at a specific location or autonomously return to a designated recovery point, which can be fixed or mobile. In extreme weather conditions or special situations (such as meteorological disasters), it is possible to continue reducing balloon pressure to carry out meteorological detection tasks during the descent phase (providing additional data on wind speed and direction compared to normal flight conditions), maximizing the utilization of a single recoverable sounding device.

Step8 : If the data volume is too large to be fully transmitted via radio communication, it can be directly read after recovery.

Once the sounding device is airborne, it can autonomously complete all flight tasks without human intervention or terminate the mission urgently according to preset procedures, while also supporting ground station commands for urgent termination or task modification.

3 Test Flight Time, Site, and Airspace

3.1 Test Flight Time

The test flight is scheduled for early July, so the analysis of the following site is also based on the meteorological conditions of early July. Trajectory simulations will be conducted before the test flight in early July. If the final decision is to conduct the test flight at the upper air meteorological observation station, the test flight will follow the routine sounding tasks at 7:15 AM each day, completing the test flight within an hour to ensure that it does not interfere with normal sounding tasks while maximizing the use of existing resources and data environment of the upper air meteorological observation station. Since the launch times are close together, the ascent flight trajectory will be similar to that of the upper air meteorological observation station.

Plan02 (Post-window): The test balloon will be launched at 08:50. At this time, the routine business data from the meteorological station has been successfully uploaded and the global exchange has concluded. The pressure, temperature, and radar reception systems have been shut down, and the test launch will not impose any time pressure on the normal operations of the on-duty personnel at the meteorological station.

3.2 Initial Screening of Test Flight Site

To facilitate the application for the airspace of the sounding balloon (which can be released at intervals with the high-altitude meteorological observation station), utilize existing infrastructure for launch, and leverage the forecasts and meteorological data on the day of launch from the high-altitude meteorological observation station, priority will be given to conducting test launches at the existing high-altitude meteorological observation stations. The situation of the 8 high-altitude meteorological observation stations in Qinghai Province is as follows.

In addition, the Lenghu Rocket Launch Base in Mangya City, at an altitude of 2777m, is located in the ruins of an abandoned town and a desert area, ensuring high recovery safety; the weather is clear with few clouds and high atmospheric transparency. Due to infrastructure and military management factors, it is considered as the second choice site after the Tuotuo River high-altitude meteorological observation station.

In summary, the preliminary screening of test launch sites includes the Tuotuo River high-altitude meteorological observation station (latitude34.215°, longitude92.440°) and the Lenghu Rocket Launch Base in Mangya City (latitude36.6188°, longitude93.3675°).

3.3 Initial Screening Flight Test Site Analysis

The Mangya high-altitude meteorological sounding station and its external space in Qinghai Province. The geographical environment of Mangya is unique, with a radius of one hundred kilometers surrounding it consisting of vast sandy deserts, salt flats, and uninhabited Gobi areas with flat surfaces and sparse population, without civilian railways, high-voltage power grids, or important civilian facilities. This creates an almost perfect natural undamaged hard landing buffer zone, ensuring zero personnel casualties and zero property damage during the landing of gliding vehicles and ground clearance.

Table 1 High-altitude meteorological observation stations in Qinghai Province

Station Name	Altitude(m)	Regional Terrain and Surrounding Environment Recovery Safety	Preliminary Screening Conclusion	
Mangya Station	3,100	On the western edge of the Qaidam Basin, surrounded by large areas of salt flats, deserts, and flat Gobi.	Relatively high, located on the edge of Mangya City, with certain safety risks.	The terrain is extremely flat, with an excellent electromagnetic environment, considered as an alternative option due to safety issues.
Delingha Station	3,200	In the eastern part of the Qaidam Basin, with flat and open desert to the south and west, and high mountains to the north.	The population is concentrated in the urban area, but the surrounding desert areas are sparsely populated.	Not considered due to terrain limitations.
Golmud Station	2,780	In the central part of the Qaidam Basin, with large areas of gravel Gobi, where transportation and military jurisdictions intersect.	Low; industry, military, transportation and densely populated.	Not considered.
Tuo Tuo River Station	4,534	The second highest sounding station in the world is located on the northern slope of the Tangula Mountains, characterized by high-altitude cold wetlands, permanent snow, and clustered river channels.	Extremely high; Tangula Mountain Town has relocated all residents, with only the Qinghai-Tibet Railway nearby and	The main selected site, Tuo Tuo River Station has a research nature.
Xining Station	2,261	Huangshui Valley, surrounded by mountains, with a high degree of urbanization in the river valley area.	Extremely low, the capital of Qinghai Province, with high population density.	Not considered.
Yushu Station	3,681	In the heart of the Sanjiangyuan, a typical high mountain gorge and high cold grass land, with dramatic undulations.	Relatively high; fewer population transportation lines.	Complex terrain, not considered.
Dari Station	3,968	On the southern slope of the Bayan Har Mountains, cold hills and wide valleys, with significant surface undulations.	Relatively high; fewer population transportation lines.	Complex terrain, not considered.
Markang Station	2300	Located in the southeastern edge of Qinghai, it is mostly mountainous and features high-altitude meadows.	Extremely high, with relatively few population traffic lines.	Complex terrain, not considered.

3.3.1 Terrain

- The Tuotuo River High-altitude Meteorological Observation Station: Located in Tanggu La Mountain Town, Golmud City, Haixi Mongolian and Tibetan Autonomous Prefecture, Qinghai Province, it sits on a flat plateau at the northern foot of the Tanggu La Mountain Range. Terrain undulation and recovery accessibility: Although the area is relatively open and flat, the base elevation is extremely high (4533 meters to 4547 meters). Early July coincides with the rainy and warm season of the Tibetan Plateau, where the active layer of permafrost melts, forming large areas of plateau marshes, wetlands, and thawed muddy zones. Vehicles can easily get stuck in the mud in the wilderness without road network coverage, which poses extremely high logistical and life safety risks for ground search and vehicle recovery if the sounding device fails to achieve precise return and lands in the wilderness.
- The Lenghu Rocket Launch Base is located in the desert Gobi at the northwestern edge of the Qaidam Basin. Terrain undulation and recovery accessibility: The surface elevation of the base is approximately 2700 meters to 2800 meters. The surrounding area features extremely complex Yadan landforms (such as the Erbo Liang Yadan group) and the steep Saishten Mountain (with a peak elevation exceeding 4200 meters). The Yadan landform area is crisscrossed with deep ravines and has severe terrain undulation, completely within a "life forbidden zone" with no cellular communication signal coverage. If the payload falls into a deep Yadan trough or the steep cliffs of Saishten Mountain, conventional vehicles and personnel can hardly reach it, making recovery extremely difficult. Local microclimate characteristics: The Lenghu area has an extremely arid temperate desert climate, characterized by cold, windy, and dry conditions year-round. In early July, the basin experiences intense sunlight, leading to strong dry convection and sandstorm weather during the day. The terrain shielding effect of Saishten Mountain results in significant local valley wind circulation, with active low-level shear lines during the transition between afternoon mountain winds and valley winds, which is very unfavorable for the stability of the balloon's ascent.

3.3.2 Wind Field

The two sites are located in the mid-latitude transition zone of the Northern Hemisphere, where the atmospheric circulation in early July exhibits distinct summer convective layer

and stratospheric reverse characteristics:

- Tropospheric westerly wind circulation: From the surface boundary layer (about 3 km) to the tropopause (about 16 km) height, due to the influence of the prevailing westerlies in the mid-latitudes, this height range is controlled by westerlies year-round. Although the intensity of the westerly jet in summer is significantly weaker than in winter, the average westerly wind speed in the mid to upper troposphere can still reach 10 m/s to 25 m/s. This means that the balloon will experience continuous and strong eastward thrust when ascending and descending through the troposphere.
- Stratospheric easterly wind circulation: As summer approaches, the temperature gradient in the stratospheric atmosphere reverses, leading to the establishment of subtropical high pressure in the stratosphere, resulting in an extremely stable prevailing easterly wind belt in the upper stratosphere from 18 km to 35 km. In early July, the easterly wind speed near 30 km altitude is usually stable at 12 m/s to 20 m/s. At this time, if the balloon remains at this altitude, it will receive a continuous westward thrust.

3.3.3 Interference and Safety of Airplane, Railway, and Highway Transportation Corridors

Table 4: Comparison of Traffic Line Interference and Anti-Interference Safety Assessment

Assessment Dimension	Tuotuo River High-altitude Meteorological Observation Station	Langyashan City Cold Lake Rocket Launch Base
Nearby Airport	Golmud Airport (GOQ): Located to the northeast of the meteorological station, the great circle geodesic distance is approximately 323 km (the road transport distance is approximately 420 km). Yushu Batang Airport (YUS) : Located to the southeast of the meteorological station, the large circle geodesic distance is approximately 453 km . All are plateau civil branch airports.	Dunhuang Mogao International Airport (DNH): Located in the northeast direction of the base, large circle geodesic distance is approximately 211 km (actual road travel distance is about 250–270 km). Haixi Mangya Airport (Huatugou Airport, HTT): Located in the southwest direction of the base, approximately 225 km in great circle geodesic distance.
Morning flight conflict(07:15–09:45)	Medium- High risk. The airspace above the weather station is located under the busy high-altitude flight routes connecting the southwestern hub (Chengdu, Chongqing) and Lhasa Gonggar Airport (such as L888 , Y95 , etc.). 08:30–09:30 is the concentrated period for flights to Tibet, such as China International Airlines CA2537 (approximately 08:30–09:15) and Sichuan Airlines 3U8691 (approximately 08:45–09:15). During this period, when the release platform ascends through 8400 m to 11600 m in the civil aviation cruising altitude layer, there is a risk of route collision or safety distance conflict.	No conflict. The base is located in an uninhabited desert area, with no terminal approach zones or low-altitude approach areas of any civil aviation airports within a radius of 200 km. During the period of 07:15–09:45, the base's takeoff and landing position has no regular civil passenger flight tracks below low altitude (6000 m). At high altitude, there are only a few transit routes, the ground airspace is open, and there are no interaction points in air traffic control, resulting in collision probability approaching zero.
Adjacent railway	Qinghai-Tibet Railway: The nearest railway line is approximately 1.5 kilometers east of the weather station. The geodesic distance from the weather station to the nearest manned railway station, Tuotuohe Station (latitude 34.2331° , longitude 92.4457°), is about 2.09 kilometers. This railway line has been electrified and operates with a high-voltage contact network (AC 25 kV).	Dungr Railway (from Dunhuang to Golmud): The railway line runs along the west side of Mangya City. 2219 National Highway, east of Golmud, approximately 76 kilometers from the Dunhuang Line Control Station. The nearest railway passenger and freight station, Mahai Station (Latitude 38.3951° , Longitude 94.5951°), is about 89.8 kilometers away. There is no physical interference risk on the ground.
Morning Train Conflicts(07:15–09:45)	Extremely high risk. There are concentrated passenger	No conflicts. There are no passenger trains operating

3.4 Airspace

3.4.1 Explanation of Airspace Situation

This device uses a standard 750g high-altitude balloon, consistent with conventional sounding in Qinghai Province, thus the ascent phase is similar to traditional sounding balloons. The descent phase is controllable, resulting in actual airspace occupancy being less than that of traditional sounding balloons, and it can actively avoid high-risk areas. If released at intervals from high-altitude meteorological observation stations, the

airspace can utilize the existing airspace of the high-altitude meteorological observation stations or apply for airspace based on the existing airspace of the high-altitude meteorological observation stations.

3.4.2 Simulation of Probe Balloon Flight Trajectory

Using publicly available high-altitude wind field data, trajectory simulations were conducted for two test flight sites on March 6, with an ascent rate of 400m/min, while simulating a powered descent at a descent rate of 400m/min. The generated KML trajectory is imported into map GS(2025)3131 as shown below.

This simulation method has been validated against the actual flight trajectory of the Xining high-altitude meteorological observation station, and the comparison results indicate a high accuracy, which can be used for trajectory prediction of sounding balloons in the Qinghai-Tibet Plateau region.

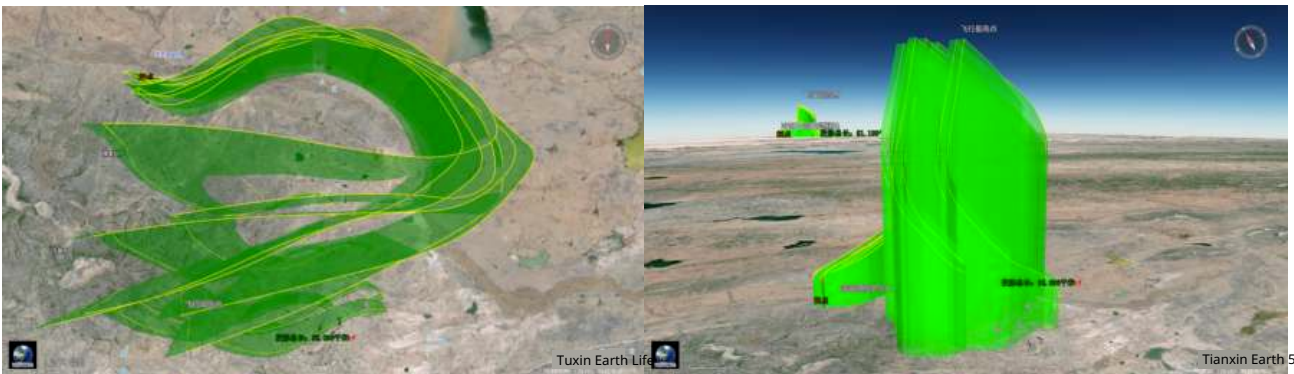


Figure 7: Tuxin River High-altitude Meteorological Observation Station Trajectory Prediction



Figure 8: Tuxin River Launch Site and Landing Point Prediction

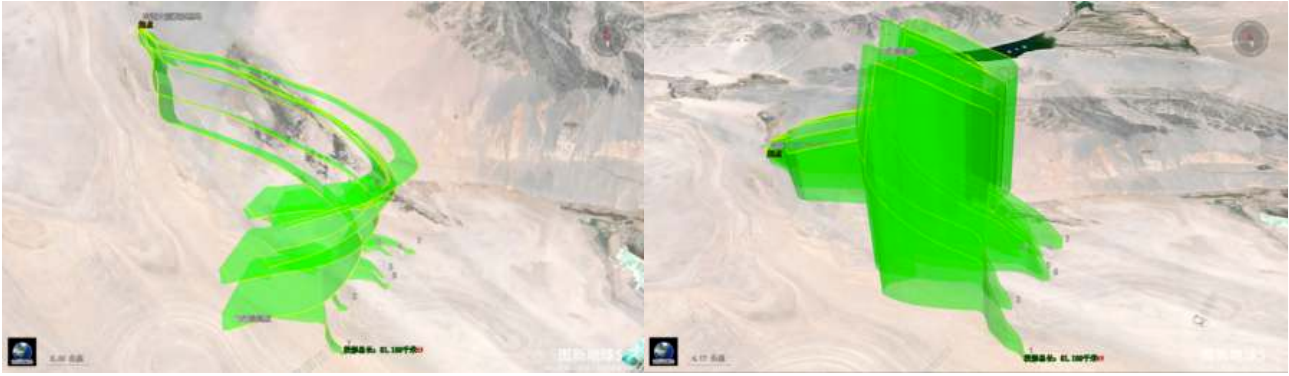


Figure19: Cold Lake Rocket Launch Base Trajectory Prediction

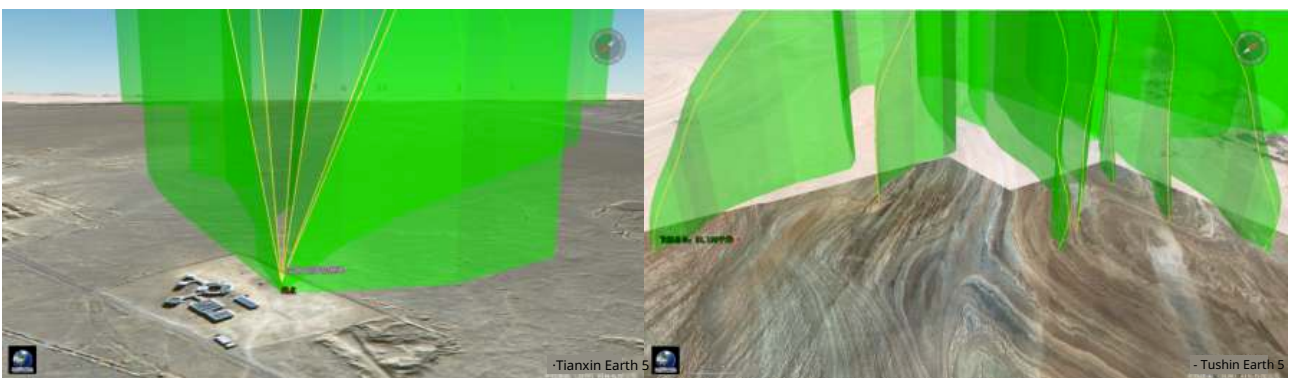


Figure20: Cold Lake Launch Site and Landing Point Prediction

Both test flight sites simulate 7 flight trajectories starting from 7:15 AM on June 3rd, Beijing time, with intervals of 3 hours. In each figure, the labels 1, 2, 3, 4, 5, 6, 7 correspond to the launch times of 7:15, 10:15, 13:15, 16:15, 19:15, 22:15, and 1:15 the next day.

The simulation results show that, influenced by the westerly wind belt, the sounding balloon drifts eastward during the ascent phase; influenced by the easterly wind belt, it drifts westward in the upper stratosphere. The general flight direction and landing area can be predicted.

Specifically, the Cold Lake Rocket Launch Base's flight is generally directed southeast, with landing points distributed in the uninhabited area southeast of Cold Lake. The entire flight path does not pass over any civil airports, railways, or populated areas, with the highest point of flight having a maximum projected distance of about 28 km from the launch site, and the landing point being about 52 km away from the launch site.

The Tuo Tuo River high-altitude meteorological observation station's flight is generally directed eastward, with landing points distributed in the uninhabited area southeast of the Tuo Tuo River station. Initially, it drifts slightly westward due to near-surface wind direction, but after ascending to 2000 m above the ground, it turns eastward. The flight only approaches the Qinghai-Tibet Railway and G109 National Highway about 1.5 km from the launch point during the initial phase, consistent with the station's routine sounding. The highest point of flight has a maximum projected distance of about 20 km from the launch site, and the landing point is about 32 km away from the launch site.

4 Test Flight Plan and Safety Assurance Scheme

The traditional sounding instrument is uncontrollable throughout the flight, with free fall during the descent phase; this device's ascent phase is consistent with traditional sounding instruments.

However, since the descent phase is controllable, the actual airspace occupied is smaller than that of traditional sounding balloons, and it can actively avoid high-risk areas.

4.1 Flight Test Preparation

If the field test flight is feasible, the production of the sounding device will be completed in June, followed by low-altitude testing. We will actively contact relevant units for high-altitude low-temperature simulation testing to ensure the device's reliability and safety in extreme high-altitude environments. We will contact manufacturers to customize low-temperature equipment, including non-lubricated low-temperature servos, imaging equipment, etc. After the atmospheric model is opened, we will promptly simulate the test flight site to optimize the launch time window and recovery plan.

4.2 Security Assurance Plan

Electronic Geofencing and Automatic Separation Free Fall Mechanism:

A three-dimensional electronic fence constructed based on the approved airspace: a buffer zone is defined within the airspace, and once the coordinates are determined to reach the buffer zone, the termination procedure is triggered:

Action One: Forced Separation: The electric push rod retracts, and the sounding device actively separates from the high-altitude balloon and enters free fall, appropriately deploying the slanted wings to glide in the opposite direction of the airspace boundary, reducing the descent speed.

Action Two: If the electric push rod freezes or the mechanical lock jams, preventing the physical separation mechanism from releasing the balloon, the solenoid valve fully opens, rapidly releasing gas to make the balloon lose lift, and the sounding device enters a free fall state.